The Penguins

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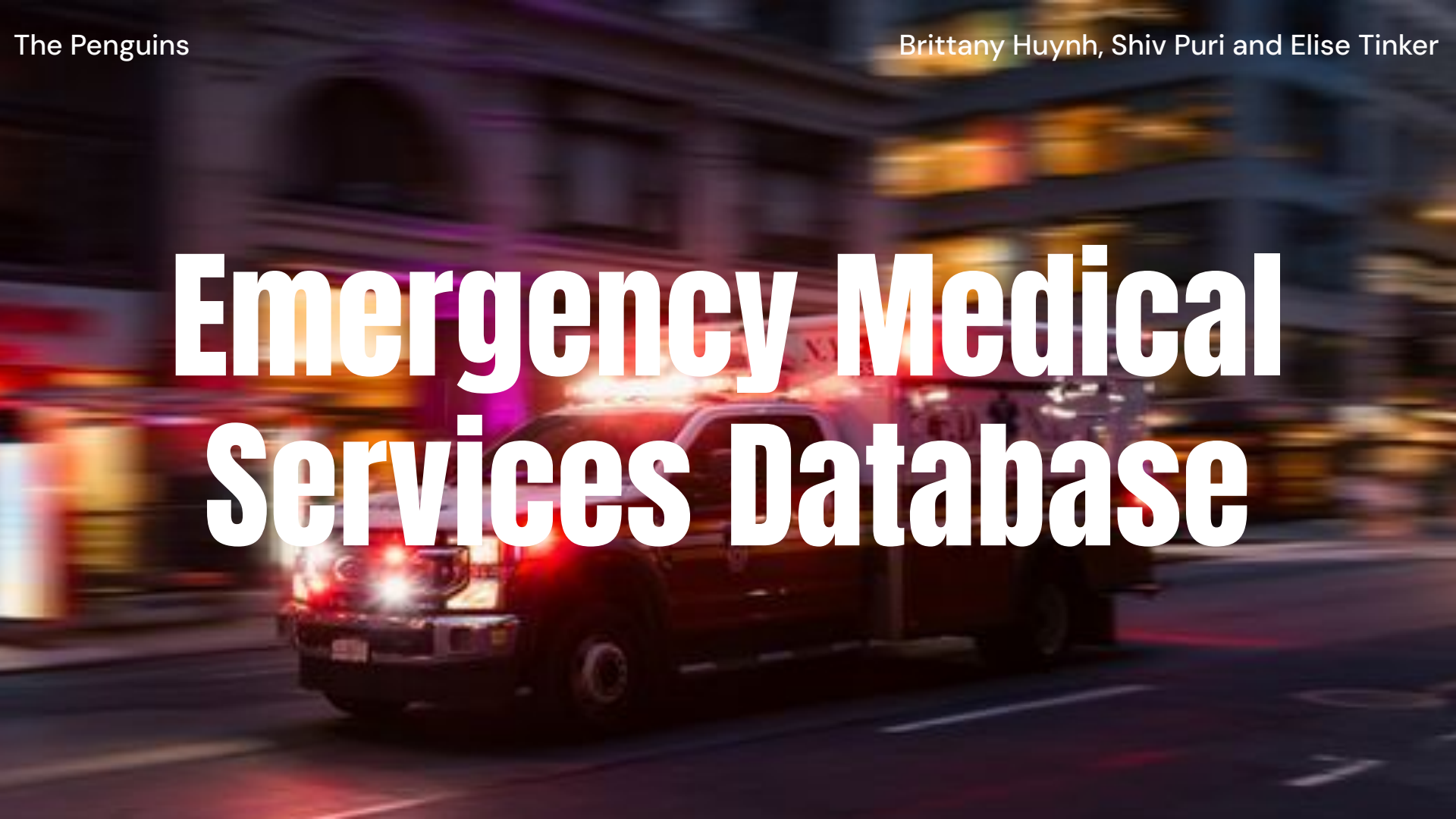
Emergency Medical Services Database



Project Description

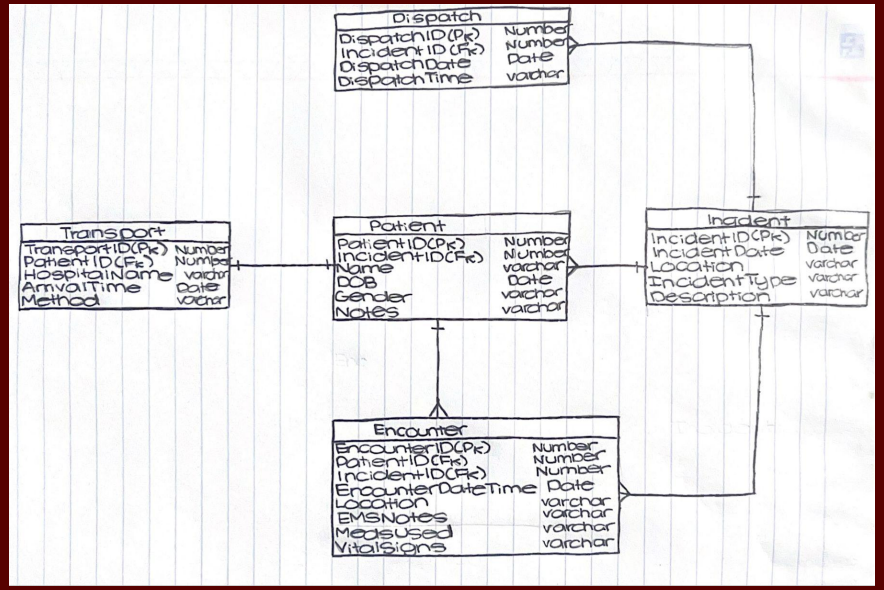
- Background
  - Emergency Medical Services (EMS) are typically the first responders that arrive on scene when an emergency occurs. In these high-pressure situations, accurate documentation is essential for patient care, decision-making, and coordination between teams. However, EMS data is often stored in separate systems or recorded manually, making it difficult to track incidents, patient history, and medical actions taken in the field.
- Use Case
  - Our database is designed to support EMS personnel during real emergency. It allows EMS agencies to quickly retrieve information, analyze patterns, and maintain accurate medical and operational records.
- Data Source
  - Our database uses fictional sample data, for the purpose of demonstration and testing purposes. No real patient information is used, however the structure of our database is designed to allow for real EMS data to be integrated in the future.

Data Model

- Table #1: Incident
  - Stores details about each emergency event (date, location, type, and description).
  - Primary Key(s): IncidentID
  - Foreign Key(s): None
  - Constraints: IDs must be unique, date can't be null.
- Table #2: Dispatch
  - Records when an EMS unit was dispatched to an incident
  - Primary Key(s): DispatchID
  - Foreign Key(s): IncidentID
  - Constraints: IDs have to unique, IncidentID must match existing incident.
- Table #3: Patient
  - Holds all data that relates to an affected individual at an incident
  - Primary Key(s): PatientID
  - Foreign Key(s): IncidentID
  - Constraints: IDs must be unique, IncidentID must match existing incident.
- Table #4: Encounter
  - Logs interactions between EMS personnel and a patient during an incident, including encounter time, notes, vitals, medications, and location.
  - Primary Key(s): EncounterID
  - Foreign Key(s): PatientID & IncidentID
  - Constraints:IDs must be unique, PatientID has to match a Patient, IncidentID must match an incident, date can't be null.
- Table #5: Transport
  - Stores hospital transport details for a patient following an encounter, including arrival time, hospital name, and transport method.
  - Primary Key(s): TransportID
  - Foreign Key(s): PatientID
  - Constraints: IDs must be unique, PatientID must match Patient.

Entity-Relationship Diagram

- Relationships:
  - Incident (one-to-many) Dispatch
  - Incident (one-to-many) Patient
  - Patient (one-to-many) Encounter
  - Incident (one-to-many) Encounter
  - Patient (one-to-one) transport



Insert Query

INCIDENTID	INCIDENTD	LOCATION	INCIDENTTYPE	DESCRIPTION
1001	17-NOV-25	123 Main St	Medical	Patient with severe chest pain
1002	17-NOV-25	25 Jane Ave	Medical	Patient with sprained ankle
1003	18-NOV-25	500 Maple Rd	Trauma	Slip and fall incident

The table shown here is called Incident, and it contains information about each 911 call; the date, location, type of incident, and a brief description.

Code:

```
INSERT INTO Incident (IncidentID, IncidentDate, Location, IncidentType, Description)
VALUES (1001, TO_DATE('2025-11-17', 'YYYY-MM-DD'), '123 Main St', 'Medical', 'Patient with severe chest pain'
);

INSERT INTO Incident (IncidentID, IncidentDate, Location, IncidentType, Description)
VALUES (1002, TO_DATE('2025-11-17', 'YYYY-MM-DD'), '25 Jane Ave', 'Medical', 'Patient with sprained ankle'
);

INSERT INTO Incident (IncidentID, IncidentDate, Location, IncidentType, Description)
VALUES (1003, TO_DATE('2025-11-18', 'YYYY-MM-DD'), '500 Maple Rd', 'Trauma', 'Slip and fall incident'
);
```

Insert Query

DISPATCHID	INCIDENTID	DISPATCHD	DISPATCH
2001	1001	17-NOV-25	14:32
2002	1002	17-NOV-25	09:10
2003	1003	18-NOV-25	19:45

The table shown here is called Dispatch, and it stores details about when EMS units were dispatched, including the dispatch date, time, and the incident they were responding to.

Code:

```
INSERT INTO Incident (IncidentID, IncidentDate, Location, IncidentType, Description)
VALUES (1001, TO_DATE('2025-11-17', 'YYYY-MM-DD'), '123 Main St', 'Medical', 'Patient with severe chest pain'
);

INSERT INTO Incident (IncidentID, IncidentDate, Location, IncidentType, Description)
VALUES (1002, TO_DATE('2025-11-17', 'YYYY-MM-DD'), '25 Jane Ave', 'Medical', 'Patient with sprained ankle'
);

INSERT INTO Incident (IncidentID, IncidentDate, Location, IncidentType, Description)
VALUES (1003, TO_DATE('2025-11-18', 'YYYY-MM-DD'), '500 Maple Rd', 'Trauma', 'Slip and fall incident'
);
```

Insert Query

INCIDENTID	INCIDENTD	LOCATION	INCIDENTTYPE	DESCRIPTION
1001	17-NOV-25	123 Main St	Medical	Patient with severe chest pain
1002	17-NOV-25	25 Jane Ave	Medical	Patient with sprained ankle
1003	18-NOV-25	500 Maple Rd	Trauma	Slip and fall incident

DISPATCHID	INCIDENTID	DISPATCHD	DISPATCH
2001	1001	17-NOV-25	14:32
2002	1002	17-NOV-25	09:10
2003	1003	18-NOV-25	19:45

PATIENTID	INCIDENTID	NAME	DOB	GENDER	NOTES
3001	1001	John Doe	10-MAY-85	Male	History of hypertension
3002	1002	Mary Smith	20-FEB-90	Female	Ankle injury
3003	1003	Alan Johnson	30-NOV-78	Male	Head trauma – unresponsive initially

ENCOUNTERID	PATIENTID	INCIDENTID	ENCOUNTER	LOCATION	EMSNOTES	MEDSUSED	VITALSIGNS
4001	3001	1001	17-NOV-25	Ambulance	Severe chest pain, ECG abnormal	Aspirin	BP: 130/85, Pulse: 100
4002	3002	1002	17-NOV-25	Scene	Swollen ankle, difficulty walking	Ibuprofen	BP: 120/80, Pulse: 88
4003	3003	1003	18-NOV-25	Ambulance	Head injury, possible concussion	None	BP: 110/70, Pulse: 95

TRANSPORTID	PATIENTID	HOSPITALNAME	ARRIVALTI	METHOD
5001	3001	Johns Hopkins Hospital	17-NOV-25	Helicopter
5002	3002	Grey Sloan Memorial Hospital	17-NOV-25	Ambulance
5003	3003	Seattle Grace Hospital	18-NOV-25	Ambulance

Select Query

NAME	HOSPITALNAME	METHOD	ARRIVALTI
John Doe	Johns Hopkins Hospital	Helicopter	17-NOV-25
Mary Smith	Grey Sloan Memorial Hospital	Ambulance	17-NOV-25
Alan Johnson	Seattle Grace Hospital	Ambulance	18-NOV-25

This Select query shows which hospitals each patient was transported too, their method of transport, and the date it occurred.

Code:

```
SELECT Patient.Name, Transport.HospitalName, Transport.Method, Transport.ArrivalTime
FROM Transport
JOIN Patient ON Transport.PatientID = Patient.PatientID;
```

Select Query

NAME	MEDSUSED
John Doe	Aspirin
Mary Smith	Ibuprofen
Alan Johnson	None

This Select Query shows which meds were used on patients.

Code:

```
SELECT Patient.Name, Encounter.MedsUsed
FROM Encounter
JOIN Patient ON Encounter.PatientID = Patient.PatientID;
```

Update Query

INCIDENTID	INCIDENTD	LOCATION	INCIDENTTYPE	DESCRIPTION
1001	17-NOV-25	123 Main St	Medical	Patient with severe chest pain
1002	17-NOV-25	25 Jane Ave	Medical	Patient with sprained ankle
1003	18-NOV-25	500 Maple Rd	Trauma	Slip and fall incident

INCIDENTID	INCIDENTD	LOCATION	INCIDENTTYPE	DESCRIPTION
1001	17-NOV-25	123 Main St	Medical	Patient with severe chest pain
1002	17-NOV-25	456 Winter Street	Medical	Patient with sprained ankle
1003	18-NOV-25	500 Maple Rd	Trauma	Slip and fall incident

This update query corrected an error made by an EMT who originally recorded the address of the incident to be ‘25 Jane Ave’. It was updated to the correct address: ‘456 Winter Street.’

Code:

```
UPDATE Incident
SET Location = '456 Updated Street'
WHERE IncidentID = 1002;
```

Update Query

DISPATCHID	INCIDENTID	DISPATCHD	DISPATCH
2001	1001	17-NOV-25	14:32
2002	1002	17-NOV-25	09:10
2003	1003	18-NOV-25	19:45

DISPATCHID	INCIDENTID	DISPATCHD	DISPATCH
2001	1001	17-NOV-25	14:20
2002	1002	17-NOV-25	09:10
2003	1003	18-NOV-25	19:45

This update query corrected an error where the dispatch had recorded an incorrect dispatch time and updated it to the accurate one.

Code:

```
UPDATE Dispatch
SET DispatchTime = '14:20'
WHERE DispatchID = 2001;
```

Aggregation Query

HOSPITALNAME	NUMBEROFPATIENTS
Seattle Grace Hospital	1
Johns Hopkins Hospital	1
Grey Sloan Memorial Hospital	1

This Aggregation Query calculates the total count of patients that are taken to different hospitals around the area.

Code:

```
SELECT Transport.HospitalName,
COUNT(Transport.TransportID) AS NumberOfPatients FROM Transport
JOIN Patient ON Transport.PatientID = Patient.PatientID
GROUP BY Transport.HospitalName;
```

Aggregation Query

INCIDENTTYPE	NUMBEROFPATIENTS
Medical	2
Trauma	1

This aggregation query counts the total number of patients for each type of incident which occured.

Code:

```
SELECT Incident.IncidentType, COUNT(Patient.PatientID) AS NumberOfPatients
FROM Incident
JOIN Patient ON Incident.IncidentID = Patient.IncidentID
GROUP BY Incident.IncidentType;
```

Join Queries

NAME	MEDSUSED	ENCOUNTER	LOCATION
Mary Smith	Ibuprofen	17-NOV-25	Scene

This join query joins the patient's name, the medication administered, the date, and where the medication was taken. This join query is performed by combining info from Patient and Encounter tables.

Code:

```
SELECT Patient.Name, Encounter.MedsUsed, Encounter.EncounterDateTime, Encounter.Location
FROM Encounter
JOIN Patient ON Encounter.PatientID = Patient.PatientID
WHERE Encounter.MedsUsed IS NOT NULL AND Encounter.Location = 'Scene';
```

Conclusion

- Lessons Learned
  - How important planning with an ER Diagram is for database design before introducing SQL code into the project
  - How constraints within SQL code can help properly sort data in a database and avoid errors.
  - Improved our skills when it comes to writing and developing SQL code
- Challenges
  - Creating a database that is comprehensive but also simple for database management
    - We overcame this by merging some of our tables together into one for our final project to reduce the quantity of tables.
  - Making sure all the foreign keys and relationships were set up correctly so that the data stayed consistent across tables.
    - We carefully followed the order of our ER diagram when writing inserts; starting with Incident, Dispatch, Patient, Encounter, then Transport. We added primary key and foreign key constraints.
- Future Work
  - Implement a user interface that provides a way for EMS personnel to enter and view system data more easily
  - Use real EMS datasets or collaborate with local agencies to test the system in practical settings

Thank you!

Any Questions?